DARSINI LAKSHMIAH

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SUMMARY

Data professional with 4+ years of experience analyzing large-scale customers and transaction data to generate product and marketing insights. Experienced in SQL, Python, experimentation, and dashboarding, with a track record of building analytics platforms and reporting systems at Amazon, AARP, and NatWest to support data-driven decisions.

EDUCATION

George Washington University, Washington, DC
Master of Science in Data Science, GPA - 4.0/4.0 **Expected May 2026**

Madras Christian College, Chennai, India
Bachelor of Science in Mathematics, CGPA - 8.50/10 **May 2021**

WORK EXPERIENCE

AARP SERVICES INCORPORATED, Washington, DC, USA

Data Scientist Intern **September 2025 – Present**

ASI Marketing Analytics Team - supporting campaign analytics, personalization, and marketing data infrastructure for AARP's digital content platform.

- **Reduced manual campaign operations by ~25% by automating ~30 recurring marketing campaigns** previously executed in SAS using n8n workflows integrated with Databricks, PySpark, and Redshift APIs, enabling scheduled execution, automated reporting, and monitoring for marketing analytics teams.
- Analyzed millions of user interaction records from S3 and Redshift to understand engagement patterns across AARP digital platforms and **developed a LightGBM-based recommendation model that improved campaign targeting** across content categories, increasing user engagement by **~10%**.
- **Built an interactive marketing analytics dashboard using Databricks and Redshift to monitor key campaign KPIs** (open rates, CTR, engagement trends), enabling marketing teams to quickly evaluate performance and optimize campaign strategy.
- **Improved accessibility of analytics insights by developing a LLaMA-powered analytics assistant** integrated with Databricks and Redshift, enabling marketing and product teams to explore campaign performance and user behavior through natural language queries and reducing analytics turnaround time by **~50%**.

AMAZON, Austin, Texas, USA

Business Intelligence Engineer Intern **May 2025 - August 2025**

Grocery Data Analytics team - Customer 360 Behavioral Analytics Platform for Amazon Grocery.

- **Enabled data-driven decisions for a $200M multi-banner grocery retail initiative** by developing a unified Customer Behavioral Analytics platform integrating online and in-store purchase data across Amazon Fresh and Whole Foods; analyzed lifetime value and purchasing patterns across millions of transactions and hundreds of store locations using Quick Sight and SQL.
- **Improved customer targeting and product experimentation capabilities** by building a purchase-propensity prediction model using Python, SQL, and Amazon SageMaker; integrated predictions into analytics dashboards used by product and retail teams to evaluate feature launches and customer engagement trends.
- **Reduced analytics latency and improved reliability of operational dashboards by ~20%** by engineering scalable ETL pipelines with AWS Glue, Athena, S3, and Redshift, transforming raw transactional data into optimized SQL aggregation tables powering large-scale retail analytics.
- **Provided actionable insights for new in-store digital features** such as Dash Cart and Click-Checkout by building operational performance dashboards and analyzing store-level adoption patterns; presented findings to Product Managers, Data Scientists, and retail operations stakeholders guiding the Amazon Fresh, Whole Foods integrated store strategy.

GEORGE WASHINGTON UNIVERSITY, Washington DC, USA

Graduate Assistant & Data Science Consultant **October 2024 - May 2025**

- Taught and supported instruction in NLP using Python, NLTK, spaCy, and Scikit-learn (text preprocessing, feature engineering, model evaluation), while developing ArcGIS Online story maps and analytical dashboards to help NGOs visualize global impact trends, forecast outcomes, and communicate data-driven insights to stakeholders.

NATWEST GROUP PRIVATE LIMITED, Chennai, India

Senior Data Analyst **Aug 2021 - June 2024**

Digital Banking Analytics & Risk Intelligence Team - building analytics platforms and models to analyze customer behavior, optimize digital campaigns, and strengthen fraud detection across UK and EMEA banking systems.

- Built distributed ETL pipelines and Redshift data models using AWS EMR (Spark, PySpark, Airflow) to process 1.5M+ daily banking transactions, improving analytics query performance by **~35%** and enabling reliable reporting for product, risk, and regulatory teams.
- **Analyzed multi-channel banking transaction data to identify behavioral patterns** and developed NLP-driven anomaly detection models using TF-IDF feature engineering and XGBoost in Python, strengthening enterprise AML and fraud monitoring systems across UK and EMEA operations.
- **Conducted A/B experimentation and customer behavior analysis** using Python (statsmodels) and SQL to evaluate personalized digital marketing campaigns, increasing engagement across UK banking segments by **~9%**.
- **Built analytics datasets and reporting dashboards used by product and risk teams** to monitor digital banking performance, campaign effectiveness, and customer activity trends across multiple banking channels.

HAMARI PAHCHAN NGO, Delhi, India

Research and Development Intern **June 2020 - September 2020**

- **Crafted interactive Power BI dashboards** by integrating cleaned survey data from Google Forms and Excel VBA, enhancing program targeting and expanding outreach by **30%.**

TECHNICAL SKILLS

- **Programming:** Python, R, SQL, SAS, PySpark
- **Databases:** Databricks**,** Apache Spark, MySQL, MongoDB
- **Data Analysis & Visualization:** Pandas, NumPy, Seaborn, QuickSight, Tableau, Power BI, Excel (Advanced), A/B Testing.
- **Machine Learning & Modeling:** Scikit-learn, TensorFlow, PyTorch
- **Cloud & Big Data Tools:** AWS (Glue, Redshift, Athena, S3, Lambda, Bedrock), Google Cloud Platform (BigQuery, Cloud Functions, Cloud Storage), Apache Airflow, dbt, Snowflake
- **Tools:** Git/GitHub, Docker, Flask, VSCode
- **Productivity & Collaboration:** Confluence, Jira, Microsoft Office (Word, Excel, PowerPoint)

PROJECTS

Data Visualization Portfolio | DataViz

- Designed and developed interactive Tableau dashboards showcasing insights from real-world datasets.
- Utilized advanced Tableau features such as calculated fields, parameters, and trend analysis to enhance data storytelling.

Instacart Market Basket Analysis | Project-link

- Analyzed 3M+ user interactions using Python, Pandas, and Spark to model user journeys and build association-rule and collaborative filtering-based recommendations, improving recommendation accuracy by 25% and reducing analysis time by 30%.

Asthma Prevalence Prediction | Project_link

- Developed and evaluated machine-learning models (SVR, Gradient Boosting, Neural Networks, GPR) on CDC PLACES data to forecast county-level asthma prevalence, achieving $R^2 = 0.79$ and enabling interpretable, policy-relevant public health insights.